

Clique and chromatic numbers of binary subspace orthogonality graphs

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Abstract

We determine the clique number of the orthogonality graph on the nonzero subspaces of $\mathbb{F}_2^n$ and establish that its chromatic number in dimension six is 15. The clique upper bound follows from a radical and quotient argument. The chromatic upper bound is certified by a complete coloring of all 2824 nonzero subspaces and an independent check against coordinate orthogonality. For the induced graph on lines, we prove a weighted independent-set bound and construct a 13-coloring, giving $12 \leq \chi(\Gamma_6) \leq 13$. Thus the full subspace graph requires at least two more colors than its line subgraph. We describe the geometry of the fixed-clique color lists and a dimension reduction for extending the search to dimension seven.

1 Introduction and statements of the results

Let $V_n = \mathbb{F}_2^n$ be equipped with the standard bilinear form

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i.$$

The *orthogonality graph* $\mathcal{O}_n^*$ has as its vertices the nonzero linear subspaces of V_n . Two distinct vertices U, W are adjacent when every vector of U is orthogonal to every vector of W . Its induced subgraph on the one-dimensional subspaces is denoted by Γ_n . Over $\mathbb{F}_2$, each line has a unique nonzero vector, so Γ_n can equivalently be described as the graph on the nonzero vectors of V_n with two distinct vectors adjacent when their dot product is zero.

The graph $\mathcal{O}_n^*$ was introduced by Nikandish [1] in connection with the annihilating-ideal graph of the ring

$$R_n = \frac{\mathbb{F}_2[x_1, \dots, x_n, z]}{(z^2, zx_i, x_i x_j (i \neq j), x_i^2 - z)}.$$

Indeed, the annihilating-ideal graph $\mathbb{AG}(R_n)$ is isomorphic to $\mathcal{O}_n$, the graph on all subspaces, including zero. The zero subspace is a universal vertex, so deleting it gives $\mathcal{O}_n^*$ and reduces both the clique and chromatic numbers by one [1, Theorem 2.4 and equations (3)–(4)]. Nikandish [1] posed the following problems.

Problem 1.1 ([1], Problem 4.2). Determine the exact value of $\omega(\mathcal{O}_n^*)$ for every n .

Problem 1.2 ([1], Problem 4.3). Determine $\chi(\mathcal{O}_n^*)$ for $n \geq 6$ and investigate the relation between $\chi(\mathcal{O}_n^*)$ and $\omega(\mathcal{O}_n^*)$.

Problem 1.3 ([1], Problem 4.4). Determine whether there exist infinitely many $n \geq 1$ with $\chi(\mathcal{O}_n^*) > \omega(\mathcal{O}_n^*)$.

Put

$$N(r) = \sum_{d=1}^r \begin{bmatrix} r \\ d \end{bmatrix}_2,$$

where $\begin{bmatrix} r \\ d \end{bmatrix}_2$ is the Gaussian binomial coefficient, and set $N(0) = 0$. Our results are

$$\omega(\mathcal{O}_n^*) = \max\{n, N(\lfloor n/2 \rfloor) + (n \bmod 2)\} \quad (n \geq 1), \quad (1)$$

and

$$\chi(\mathcal{O}_6^*) = 15, \quad 12 \leq \chi(\Gamma_6) \leq 13. \quad (2)$$

The first formula answers Problem 1.1 in all dimensions. The second answers the dimension-six case of Problem 1.2 and quantifies the separation between coloring the line graph and coloring the full subspace graph.

The clique argument is algebraic: it sums the radicals of an arbitrary clique and passes to the resulting orthogonal quotient. The full-graph chromatic upper bound uses an explicit finite certificate, while the line lower bound follows from the geometry of pairwise nonorthogonal vectors. The verification details and the scope of the Lean formalization are collected in Section 7.

The paper is organized as follows. Section 2 contains the proof of (1). Section 3 treats the line graph Γ_6 and establishes the bounds in (2). Section 4 proves $\chi(\mathcal{O}_6^*) = 15$. Section 5 describes the geometry of the precoloring. Section 6 records the dimension reduction for $n = 7$ and states the remaining open problems. Section 7 collects the computational certificates and the Lean formalization.

2 The exact clique number

Let $N(r)$ denote the number of nonzero subspaces of $\mathbb{F}_2^r$, with $N(0) = 0$. Equivalently, $N(r) = \sum_{d=1}^r \begin{bmatrix} r \\ d \end{bmatrix}_2$.

Theorem 2.1. *For every positive integer n ,*

$$\omega(\mathcal{O}_n^*) = \max\{n, N(\lfloor n/2 \rfloor) + (n \bmod 2)\}.$$

Proof. We prove the upper and lower bounds separately.

Lower bound. The n coordinate lines $\langle e_1 \rangle, \dots, \langle e_n \rangle$ are pairwise orthogonal and form a clique of size n . For the second term, let $r = \lfloor n/2 \rfloor$ and take

$$T = \text{span}\{e_1 + e_2, e_3 + e_4, \dots, e_{2r-1} + e_{2r}\}.$$

The form $\langle \cdot, \cdot \rangle$ vanishes on T , so every nonzero subspace of T is orthogonal to every other one. Hence the nonzero subspaces of T form a clique of size $N(r)$. When n is odd, $\dim T^\perp = r + 1$, so $T^\perp$ is a distinct additional vertex orthogonal to every subspace of T . These two constructions attain both terms of the maximum. They extend the coordinate and isotropic constructions in [1, Sections 3–4].

Upper bound. Let $\mathcal{C}$ be a clique in $\mathcal{O}_n^*$. For each $U \in \mathcal{C}$ put

$$\text{rad}(U) = U \cap U^\perp,$$

and define

$$R = \sum_{U \in \mathcal{C}} \text{rad}(U), \quad r = \dim R.$$

By construction, $R \subseteq U^\perp$ for every $U \in \mathcal{C}$, and the summands are pairwise orthogonal. Hence $R \subseteq R^\perp$, so R is totally isotropic and $2r \leq n$. Moreover,

$$U \subseteq R^\perp, \quad R \subseteq R^\perp, \quad U \cap R = \text{rad}(U) \quad (U \in \mathcal{C}).$$

For the last equality, one inclusion is the definition of R ; the other follows from $R \subseteq U^\perp$.

Since the standard dot form is nondegenerate, $\dim(R^\perp/R) = n - 2r$. The form descends to the quotient $Q = R^\perp/R$. Write $\bar{U}$ for the image of U in Q . We claim that the restricted form on $\bar{U}$ is nondegenerate. Indeed, if $u \in U$ represents a vector annihilating $\bar{U}$, then $u \in \text{rad}(U) \subseteq R$, so its image in Q is zero.

The spaces $\bar{U}$ are pairwise orthogonal in Q . Their nonzero members form an indexed direct sum: in any relation $\sum_U x_U = 0$ with $x_U \in \bar{U}$, pairing with arbitrary vectors of one $\bar{W}$ and using the nondegeneracy of the restriction to $\bar{W}$ gives $x_W = 0$. Hence there are at most $\dim Q = n - 2r$ nonzero images.

The members of $\mathcal{C}$ with zero image are precisely those contained in R , and there are at most $N(r)$ of them. Therefore

$$|\mathcal{C}| \leq N(r) + n - 2r. \quad (3)$$

For $r = 0$, this gives $|\mathcal{C}| \leq n$. For every $s \geq 1$, embedding $\mathbb{F}_2^s$ as a hyperplane in $\mathbb{F}_2^{s+1}$ gives

$$N(s+1) \geq N(s) + 2,$$

because an outside line and the whole ambient space are two distinct additional subspaces. Hence $N(s) - 2s$ is nondecreasing for $s \geq 1$. If $r \geq 1$, putting $t = \lfloor n/2 \rfloor$ in (3) gives

$$|\mathcal{C}| \leq N(t) + n - 2t = N(t) + (n \bmod 2).$$

This completes the proof of the upper bound and hence of the theorem. $\square$

Corollary 2.2. *For $n = 6$,*

$$\omega(\mathcal{O}_6^*) = N(3) = 15.$$

An explicit clique of size 15 is given by the nonzero subspaces of $T = \text{span}\{e_1 + e_2, e_3 + e_4, e_5 + e_6\}$, which is a totally isotropic 3-space.

3 The line graph in dimension six

Lemma 3.1. *An independent set in Γ_6 containing t even-weight vectors and o odd-weight vectors satisfies $3t + 4o \leq 19$. Its cardinality is at most six; cardinality six requires $(t, o) = (5, 1)$.*

Proof. Write $j = (1, 1, 1, 1, 1, 1)$. The even-weight vectors lie in $j^\perp$, whose restricted form has radical $\text{span}(j)$ and rank four. If $o = 0$, the Gram matrix of the selected vectors is $J_t + I_t$, of rank t for even t and $t - 1$ for odd t . Thus $t \leq 5$.

If $o > 0$, fix an odd selected vector a . For each even selected vector e , the vector $z_e = e + a$ belongs to $a^\perp$. These vectors are orthonormal and span a nondegenerate t -space A . The ambient space $a^\perp$ is nondegenerate of dimension five. The span D of the vectors $u + a$, over odd selected vectors u , is totally isotropic and orthogonal to A inside $a^\perp$. Hence $2 \dim D \leq 5 - t$. The odd selected vectors are distinct members of $a + D$, so $o \leq 2^{\dim D}$. Consequently $o \leq 4$ for $t = 0, 1$, $o \leq 2$ for $t = 2, 3$, and $o \leq 1$ for $t = 4, 5$. These bounds prove both claims. $\square$

Theorem 3.2. $12 \leq \chi(\Gamma_6) \leq 13$.

Proof. There are 31 nonzero even-weight and 32 odd-weight vectors. Their total weight, using weights three and four respectively, is 221. By Lemma 3.1, every color class has weight at most 19, giving $\chi(\Gamma_6) \geq \lceil 221/19 \rceil = 12$. For the upper bound, interpret an integer x as the vector whose coordinate i is its i th binary digit. The certificate `results/line-coloring.json` partitions the 63 nonzero vectors into 13 classes. The independent checker verifies that every pair of distinct vectors in each class has dot product one. $\square$

For completeness, the thirteen color classes, in the same binary integer coordinates, are displayed below. They partition $\{1, \dots, 63\}$, and their within-class dot products can be checked directly.

$$\begin{aligned} &\{3, 29, 45, 53, 57, 62\}, \quad \{12, 20, 39, 56, 59\}, \quad \{15, 18, 43, 52, 61\}, \\ &\{23, 24, 44, 47, 48\}, \quad \{16, 50, 51, 54, 58\}, \quad \{32, 46, 55, 60\}, \\ &\{13, 22, 42, 49, 63\}, \quad \{5, 19, 28, 38, 41\}, \quad \{8, 10, 25, 40\}, \\ &\{2, 26, 30, 35\}, \quad \{1, 7, 9, 17, 33\}, \quad \{6, 11, 27, 34, 37\}, \quad \{4, 14, 21, 31, 36\}. \end{aligned}$$

Any hypothetical 12-coloring has a class of size six. By the lemma it consists of an odd vector a and five even vectors e . The vectors a and $e + a$ form an orthonormal basis. An isometry therefore maps this class to $\{1, 3, 5, 9, 17, 33\}$. Fixing this class reduces the remaining question to an 11-coloring of the induced graph on the other 57 vectors.

4 The full chromatic number in dimension six

Theorem 4.1. $\chi(\mathcal{O}_6^*) = 15$.

Proof. Let $T = \text{span}(3, 12, 48)$ in binary integer coordinates, with bit $i - 1$ representing the coefficient of e_i . Thus its generators are $e_1 + e_2$, $e_3 + e_4$, $e_5 + e_6$. The form vanishes on T , so its seven lines, seven planes and T itself form a 15-clique. Hence $\chi(\mathcal{O}_6^*) \geq 15$.

For the upper bound, the file `results/full-15-coloring.json` gives a basis and a color in $\{0, \dots, 14\}$ for every nonzero subspace. The independent checker reconstructs each span and verifies basis independence, distinctness, and the dimension counts

$$63, 651, 1395, 651, 63, 1.$$

These equal the Gaussian binomial coefficients for $d = 1, \dots, 6$, which proves completeness of the catalogue. For all 3 986 076 pairs of distinct entries the checker tests orthogonality on the basis vectors. Bilinearity identifies this test with graph adjacency. All 44 968 orthogonality edges have distinct endpoint colors. This establishes the upper bound $\chi(\mathcal{O}_6^*) \leq 15$, and therefore equality. $\square$

Remark 4.2. To find the certificate, fix distinct colors on the 15 nonzero subspaces W of T . Every 15-coloring uses all colors on this clique, so relabeling colors makes this precoloring an equivalent instance. The forbidden colors for a vertex $U \not\subseteq T$ are exactly those $W \subseteq H(U) = U^\perp \cap T$. If $H(U) = T$, then $U \subseteq T^\perp = T$, a contradiction. Thus $\dim H(U) \in \{0, 1, 2\}$, giving lists of sizes

$$15 - N(\dim H(U)) \in \{15, 14, 11\}.$$

Remark 4.3. A vertex whose residual degree is smaller than its list size can be deleted and restored greedily after the remaining graph is colored. Repeated deletion removes 729 uncolored vertices and leaves 2 080 vertices with 39 400 edges. The list-coloring encoding has 28 600 variables and 627 510 clauses. There is one Boolean variable for each allowed vertex-color pair. Each vertex

has an at-least-one clause and pairwise at-most-one clauses; each edge forbids simultaneous use of any color common to its endpoints' lists. Hence satisfying assignments are exactly the proper list colorings of the residual graph. A satisfying assignment was extended in reverse deletion order to produce the full certificate checked above.

5 Geometry of the precoloring

In this section we describe the geometric color lists and profiles of the 15-coloring constructed in Section 4. Keep the notation of that section: T is the totally isotropic 3-space

$$T = \text{span}\{e_1 + e_2, e_3 + e_4, e_5 + e_6\},$$

$\mathcal{C}$ is the 15-clique formed by the nonzero subspaces of T , and each $W \in \mathcal{C}$ carries its own color c_W . For a vertex U of $\mathcal{O}_6^*$ outside $\mathcal{C}$, define

$$H(U) = U^\perp \cap T, \quad K(U) = U \cap T.$$

Since $T^\perp = T$ and $U \not\subseteq T$, we have $H(U) \neq T$, so $\dim H(U) \in \{0, 1, 2\}$. The following observation is the key to the precoloring.

Lemma 5.1. *Let U be a vertex of $\mathcal{O}_6^*$ outside $\mathcal{C}$ and let $W \in \mathcal{C}$. Then U is adjacent to W if and only if $W \subseteq H(U)$. Consequently, the allowed color list of U is*

$$L(U) = \{c_W : 0 \neq W \leq T, W \not\subseteq H(U)\},$$

and its size is

$$|L(U)| = \begin{cases} 15, & \dim H(U) = 0, \\ 14, & \dim H(U) = 1, \\ 11, & \dim H(U) = 2. \end{cases}$$

Proof. By definition, U is adjacent to W if and only if $W \subseteq U^\perp$. Since $W \subseteq T$, this is equivalent to $W \subseteq U^\perp \cap T = H(U)$. The list size is $15 - N(\dim H(U))$, where $N(0) = 0$, $N(1) = 1$, and $N(2) = 4$; this gives 15, 14, and 11 in the three cases. $\square$

Table 1 groups the $2809 = 2824 - 15$ vertices outside $\mathcal{C}$ by the dimensions of U , $H(U)$, and $K(U)$.

The table has 12 dimension profiles. Its 357 vertices with $\dim H(U) = 2$ are the most constrained, having 11 allowed colors. These lists describe the constraints imposed by the clique. The full certificate in Section 4 establishes that the precoloring extends by checking every orthogonality edge, including those between vertices outside the clique.

5.1 Towards a structural description

Retaining the actual subspaces $H(U)$ and $K(U)$, rather than their dimensions alone, gives the exact profiles $(\dim U, H(U), K(U))$. Grouping the archived coloring by this triple produces 240 profiles outside the clique; 139 receive more than one color and the remaining 101 receive a single color. Thus even the exact profile does not determine the color in this fixed certificate. These statistics are recorded in `results/coloring-geometry.json` and reproduced by `develop/analyze_geometry.py`. The statistics alone do not settle whether a different proper coloring could depend on these profiles. The following obstruction rules out such a rule even with the quotient-form isometry type added.

$\dim U$	$\dim H(U)$	$\dim K(U)$	Vertices
1	2	0	56
2	1	0	448
2	2	1	196
3	0	0	512
3	1	1	784
3	2	2	98
4	0	1	448
4	1	2	196
4	2	3	7
5	0	2	56
5	1	3	7
6	0	3	1

Table 1: Distribution of the 2809 vertices outside the 15-clique, according to the triple $(\dim U, \dim H(U), \dim K(U))$.

Proposition 5.2. *No proper coloring of $\mathcal{O}_6^*$ depends solely on $(\dim U, H(U), K(U))$ and the isometry type of the restricted nondegenerate bilinear form on $U/(U \cap U^\perp)$.*

Proof. The distinct lines $U = \text{span}(e_1)$ and $V = \text{span}(e_2)$ are adjacent. Both have dimension one, $H(U) = H(V) = \text{span}(e_3 + e_4, e_5 + e_6)$, and $K(U) = K(V) = \{0\}$. Their restricted forms have Gram matrix [1], so their radicals are zero and their quotients are isometric. A rule depending only on the stated data assigns equal colors to these adjacent vertices, a contradiction. $\square$

The computation in `results/geometry-profile-audit.json` retains the exact subspaces: adding quotient dimension gives 263 profiles, 160 multicolored; adding alternating status makes no further split. None of the original 139 multicolored profiles is completely resolved. These are finite statistics of the fixed witness, distinct from Proposition 5.2.

Let G be the stabilizer of T in the orthogonal group of V_6 , acting on clique color labels by $g \cdot c_W = c_{gW}$.

Corollary 5.3. *No proper 15-coloring with the prescribed clique colors satisfies $c(gU) = g \cdot c(U)$ for every $g \in G$.*

Proof. The coordinate transposition $(1, 2)$ is orthogonal and fixes every vector of T , hence every clique subspace and every color label. It exchanges the adjacent lines $\text{span}(e_1)$ and $\text{span}(e_2)$. Equivariance would give these lines the same color, a contradiction. $\square$

5.2 Auxiliary choices and symmetry breaking

Proposition 5.2 and Corollary 5.3 show that a proper coloring cannot depend solely on the specified exact profile and quotient isometry type, and that a proper 15-coloring cannot be equivariant under the full stabilizer of T with its induced clique-label action. A structural description must distinguish vertices identified by these data and break the obstructing symmetry. Choosing auxiliary data is one way to do so.

A concrete example is a complement S to T in V_6 , so that $V_6 = T \oplus S$. The choice

$$S = \text{span}\{e_1, e_3, e_5\}$$

is a complement because the generators of T together with those of S form a basis of V_6 . It distinguishes the adjacent lines $\text{span}(e_1)$ and $\text{span}(e_2)$, since

$$\dim(\text{span}(e_1) \cap S) = 1 \quad \text{while} \quad \dim(\text{span}(e_2) \cap S) = 0.$$

Thus this fixed auxiliary choice breaks the coordinate swap responsible for Corollary 5.3.

A direct check on the certificate shows that the two adjacent lines receive distinct colors, 10 for $\text{span}(e_1)$ and 9 for $\text{span}(e_2)$, while every geometric quantity printed by the diagnostic `develop/file_separator.py` is identical. The output and the complement check are recorded in `results/adjacent-line-diagnostic.json`. The auxiliary datum $\dim(U \cap S)$ separates this pair; whether it explains further color choices in the certificate is a separate question.

Whether such complement-based data support a complete structural 15-coloring remains open, as a possible approach to Problem 6.2 in Section 6.

Two further directions remain for a structural description.

1. *Enrichment of the invariants.* To recover the current coloring, an additional invariant must distinguish vertices within the multi-color exact profiles. Proposition 5.2 shows that the isometry type of the nondegenerate quotient is still insufficient when combined with the exact profile. A finer rule must distinguish the placement of vertices such as the two lines in that proof. The subspaces $H(U)$ and $K(U)$ have a relative position in T , but neither is assumed to contain the other. For example, in binary coordinates $U = \text{span}(1, 3)$ gives $K(U) = \text{span}(3)$ and $H(U) = \text{span}(12, 48)$.
2. *Restricted symmetry or auxiliary choices.* Corollary 5.3 rules out equivariance under the full stabilizer, even when clique color labels are permuted. A structural construction must break some of this symmetry, for example by choosing auxiliary coordinate data such as the complement S above, or restricting to a smaller subgroup. No such construction is asserted here.

6 Higher dimensions and open problems

Proposition 6.1. *Let K be a fixed k -clique in $\mathcal{O}_n^*$. If $N(n - d) < k$, every uncolored d -dimensional vertex can be deleted by the degree-versus-list rule. Consequently, k -colorability of the residual instance after such deletions is equivalent to k -colorability of the full graph.*

Proof. For a d -space U , the orthogonal complement has dimension $n - d$. Every neighbor of U is a nonzero subspace of $U^\perp$, so $\deg U \leq N(n - d)$. If f fixed clique vertices are adjacent to U , then its residual degree is at most $N(n - d) - f$, while its list size is $k - f$. The stated inequality therefore guarantees an available color on restoring U . Apply this argument in reverse deletion order. $\square$

For $n = 7$, the nonzero subspaces of $T = \text{span}(3, 12, 48)$ together with $\text{span}(64)$ give a 16-clique. Since $N(3) = 15 < 16$, Proposition 6.1 removes all 14 606 vertices of dimension at least four from the 29 211-vertex graph. The remaining uncolored instance has 14 589 vertices and 953 057 edges. The direct pairwise list-color encoding would have 214 355 variables and 12 984 679 clauses. These exact instance sizes guide further geometric reduction and resource-bounded search. They are measured preprocessing counts; no 16-coloring or checked nonexistence certificate for dimension seven is supplied.

Problem 4.3 remains open for $n \geq 7$, as does Problem 4.4 of [1]. Within dimension six, the exact value of $\chi(\Gamma_6)$ and a structural description of an optimal full-subspace coloring are the immediate questions.

Problem 6.2. Give a purely structural proof that $\chi(\mathcal{O}_6^*) = 15$, avoiding the finite computation.

Problem 6.3. Determine the exact value of $\chi(\Gamma_6)$. It is known that $12 \leq \chi(\Gamma_6) \leq 13$.

7 Computational certificates and formal verification

The positive coloring certificates are checked using integer arithmetic and the coordinate definition of orthogonality. The full checker imports neither the search program nor a SAT solver. It establishes enumeration completeness from distinctness and Gaussian counts, then checks every pair of vertices. Three deliberately invalid certificates, respectively omitting a vertex, duplicating a subspace, and identifying the colors of adjacent coordinate lines, are rejected. The line-coloring checker independently verifies its partition and all within-class pairs.

The radical/quotient clique theorem has an existing Lean formalization. Its source is `formal/NikandishClique.lean`. The theorem is `NikandishClique.result`, inside the namespace

`D5.S3.Combinatorics.Orthogonality.`

The upstream revision and source hash are recorded in `formal/UPSTREAM.json`; dependencies must be obtained from that pinned upstream checkout.

The six-dimensional development uses vectors in $\text{Fin}(64)$, XOR addition, and the coordinate dot product modulo two. A vertex is a zero-containing, XOR-closed subset other than $\{0\}$, exactly the nonzero binary subspaces in this representation. The formal graph is simple and uses coordinate orthogonality between distinct such subsets. The catalogue contains 2825 entries including zero and 180800 adjoining-vector witnesses. The theorem `Catalogue.complete` derives coverage from the zero entry and closure under adjoining any vector; it does not assume the Gaussian enumeration counts. The theorem `Catalogue.colorable` transfers the checked catalogue coloring to this graph. A checked 15-clique supplies the converse inequality in `BinaryOrthogonality.chromatic_number_six`.

The archived Lean 4.33.0 development verifies four finite assertions with `native_decide`. The final theorem depends on the standard axioms `propext`, `Classical.choice`, and `Quot.sound`, together with four native-evaluation axioms; its computational trust therefore includes native evaluation. The generic completeness and colorability lemmas depend only on the three standard axioms. The recovered original log corrects the incomplete historical summary; see `notes/formal-verification-audit.md`. The exhaustive integer certificate independently verifies the same coloring. The weighted line bound, geometric profile counts and dimension-seven reduction are not covered by this Lean theorem.

The public package [2] contains the literal witnesses, independent checkers, search records, and SHA-256 bindings. The new geometric profiler checks every assigned color against its $H(U)$ -determined list. The dimension-seven reduction is also replayed in dimension six, where it reproduces the established core size and Boolean encoding counts.

References

- [1] R. Nikandish, *Annihilating-Ideal Graphs and Orthogonality Graphs over $\mathbb{F}_2$* , arXiv:2609.22769v1, 2026.
- [2] H. Ma, R. Nikandish and W. Zhang, *Binary subspace orthogonality graphs: research and reproducibility workspace*, <https://github.com/the-omega-institute/binary-subspace-orthogonality>.